

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0215

Trier 100 kV
April 30, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0215. The Interconnection Customer proposes a 60.5 MW solar photovoltaic facility interconnecting at the Trier 100 kV substation in the Indiana area. The request was received on March 20, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 163 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,982,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Dekalb - Richland 500 kV Line, the Hillcrest - Wayne Trace 100 kV L loads to 151.7% of its 88 MVA rating. Project DFAX: 5.8%.

Aep Pettit Avenue - Allen 100 kV: 146.5% loading (119 MVA) following the outage of the Segs 5 Substation - Hillcrest 100 kV; DFAX = 7.2%.

Wayne Trace - Allen 100 kV: 145.1% loading (112 MVA) following the outage of the Segs 5 - Hillcrest 100 kV line; DFAX = 7.2%.

The Monte Del Estado 6303 - Harper 100 kV is overloaded to 135.4% of its 145 MVA emergency rating for the loss of the Richland - Dekalb 500 kV. The project contributes 6.4% DFAX (3.8 MW) to this overload.

For the loss of Mason St 500 kV/230 kV Autotransformer, the Herald-Twin Cities - Highgate 230 kV Line loads to 132.7% of its 518 MVA rating. Project DFAX: 6.7%.

Barret Substation - Roseau County 100 kV: 132.1% loading (132 MVA) following the outage of the Monte Del Estado 6303 - Harper 100 kV; DFAX = 7.1%.

The Hillcrest - Muldoon Mill 100 kV is overloaded to 131.4% of its 117 MVA emergency rating for the loss of the Darlington Rd - Wayne Trace 100 kV. The project contributes 8.5% DFAX (5.1 MW) to this overload.

For the loss of Eufaula - Beaver 500 kV L, the Joso - Onancock Diesel Generation Facility 230 kV loads to 128.7% of its 795 MVA rating. Project DFAX: 14.6%.

The Pickford - Adams 230 kV is overloaded to 126.9% of its 531 MVA emergency rating for the loss of the Beaver - Trego 500 kV line. The project contributes 8.2% DFAX (5.0 MW) to this overload.

The Segs 5 - State Street 100 kV is overloaded to 125.6% of its 129 MVA emergency rating for the loss of the Dekalb Substation - Richland Substation 500 kV. The project contributes 8.3% DFAX (5.0 MW) to this overload.

The Weedpatch - Canal Road 230 kV Line is overloaded to 124.7% of its 513 MVA emergency rating for the loss of the Trego Substation - Beaver 500 kV line. The project contributes 8.4% DFAX (5.1 MW) to this overload.

For the loss of E Methuen - Lonetree 500 kV, the St. Lawrence - James De Young 230 kV loads to 123.8% of its 578 MVA rating. Project DFAX: 10.0%.

For the loss of Trego - Beaver 500 kV Line, the Post Rock - Newport Tie 230 kV loads to 123.3% of its 864 MVA rating. Project DFAX: 14.2%.

For the loss of Richland - Dekalb 500 kV, the Barret - Point A 100 kV loads to 123.1% of its 135 MVA rating. Project DFAX: 5.2%.

The E Methuen 500 kV/230 kV Transformer is overloaded to 120.0% of its 981 MVA emergency rating for the loss of the Swan Vly - Trego 500 kV. The project contributes 17.6% DFAX (10.6 MW) to this overload.

The Adams - Onancock Diesel Generation Facility 230 kV is overloaded to 119.7% of its 1,112 MVA emergency rating for the loss of the Trego - Beaver 500 kV. The project contributes 15.8% DFAX (9.5 MW) to this overload.

For the loss of Trego - Beaver 500 kV line, the Post Rock - Myrtle Point 230 kV L loads to 119.4% of its 931 MVA rating. Project DFAX: 13.7%.

For the loss of Richland - Dekalb 500 kV, the Pt A - Lake Marion 100 kV line loads to 117.9% of its 127 MVA rating. Project DFAX: 5.2%.

The Mone Del Estado 6303 - Sage 100 kV is overloaded to 117.2% of its 127 MVA emergency rating for the loss of the Dekalb - Richland 500 kV. The project contributes 6.1% DFAX (3.7 MW) to this overload.

The Peavne - Segs 5 100 kV is overloaded to 115.8% of its 117 MVA emergency rating for the loss of the Dekalb - Richland 500 kV. The project contributes 7.2% DFAX (4.3 MW) to this overload.

For the loss of Beaver - Eufaula 500 kV, the Onancock Diesel Generation Facility Substation - Beaver 230 kV loads to 115.2% of its 911 MVA rating. Project DFAX: 11.7%.

The Sage - Woodburn 100 kV is overloaded to 113.7% of its 119 MVA emergency rating for the loss of the Dekalb - Richland 500 kV. The project contributes 6.1% DFAX (3.7 MW) to this overload.

For the loss of Dekalb - Richland 500 kV Line, the Monroeville - Woodburn 100 kV loads to 112.2% of its 111 MVA rating. Project DFAX: 6.1%.

Darlington Road Substation - Muskogee 100 kV line: 112.1% loading (179 MVA) following the outage of the Richland - Dekalb 500 kV line; DFAX = 9.3%.

Adams - New Departure 230 kV: 111.9% loading (613 MVA) following the outage of the Trego - Beaver 500 kV; DFAX = 7.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hillcrest - Wayne Trace 100 kV Line	Rebuild line with higher-capacity conductor	\$1,445,000	13% (\$192,000)
Aep Pettit Avenue - Allen 100 kV	Rebuild line with higher-capacity conductor	\$5,588,000	13% (\$713,000)
Wayne Trace - Allen 100 kV	Rebuild line with higher-capacity conductor	\$4,018,000	13% (\$512,000)
Monte Del Estado 6303 - Harper 100 kV	Rebuild line with higher-capacity conductor	\$7,168,000	100% (\$7,168,000)
Highgate - Herald-Twin Cities 230 kV line	Rebuild line with higher-capacity conductor	\$43,215,000	2% (\$631,000)
Barret Substation - Roseau County 100 kV	Rebuild line with higher-capacity conductor	\$6,118,000	10% (\$590,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hillcrest - Muldoon Mill 100 kV	Rebuild line with higher-capacity conductor	\$4,841,000	13% (\$621,000)
Joso - Onancock Diesel Generation Facility 230 kV	Rebuild line with higher-capacity conductor	\$29,958,000	1% (\$340,000)
Pickford - Adams 230 kV	Rebuild line with higher-capacity conductor	\$18,547,000	1% (\$209,000)
Segs 5 - State Street 100 kV	Rebuild line with higher-capacity conductor	\$3,316,000	100% (\$3,316,000)
Weedpatch - Canal Road 230 kV Line	Rebuild line with higher-capacity conductor	\$14,780,000	1% (\$169,000)
James De Young - St. Lawrence 230 kV	Rebuild line with higher-capacity conductor	\$79,803,000	1% (\$740,000)
Post Rock - Newport Tie 230 kV	Rebuild line with higher-capacity conductor	\$28,121,000	1% (\$321,000)
Point A - Barret Substation 100 kV	Rebuild line with higher-capacity conductor	\$15,133,000	12% (\$1,786,000)
East Methuen 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$19,831,000	1% (\$201,000)
Adams - Onancock Diesel Generation Facility 230 kV	Rebuild line with higher-capacity conductor	\$55,158,000	1% (\$618,000)
Post Rock - Myrtle Point 230 kV L	Rebuild line with higher-capacity conductor	\$38,533,000	1% (\$441,000)
Pt A - Lake Marion 100 kV line	Rebuild line with higher-capacity conductor	\$3,739,000	12% (\$441,000)
Mone Del Estado 6303 - Sage 100 kV	Rebuild line with higher-capacity conductor	\$2,842,000	14% (\$392,000)
Peavine - Segs 5 100 kV	Rebuild line with higher-capacity conductor	\$2,853,000	7% (\$194,000)
Onancock Diesel Generation Facility Substation - Beaver 230 kV	Rebuild line with higher-capacity conductor	\$25,941,000	1% (\$300,000)
Sage - Woodburn 100 kV	Rebuild line with higher-capacity conductor	\$2,711,000	14% (\$374,000)
Monroeville - Woodburn 100 kV	Rebuild line with higher-capacity conductor	\$7,029,000	14% (\$970,000)
Darlington Road Substation - Muskogee 100 kV line	Rebuild line with higher-capacity conductor	\$15,627,000	3% (\$403,000)
Adams - New Departure 230 kV	Rebuild line with higher-capacity conductor	\$29,406,000	1% (\$340,000)
	Total allocated to project		\$21,982,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.